V-T-276H
October 19, 1976
SUPERSEDING
Fed. Spec. V-T-276G
October 15, 1970

FEDERAL SPECIFICATION

THREAD, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for cotton thread used for machine and hand sewing.

1.2 Classification.

1.2.1 Types and finishes. The thread shall be of the following types, finishes, and color categories as specified (see 6.1):

Type I - Machine thread.

IA - Soft finish.

IA1 - Unbleached.
IA2 - White and tints.
IA3 - Black and colors.

IB - Glazed finish.

IB1 - Unbleached.
IB2 - White and tints.
IB3 - Black and colors.

IC - Mercerized finish.

IC1 - White and tints.
IC2 - Black and colors.

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Type II - Basting thread.

IIA - Glazed finish.

IIA1 - White.
IIA2 - Black.

Type III - Heavy thread (unbleached, white or in colors, as specified).

IIIA - Soft finish.
IIIB - Glazed finish.

Type IV - Shoe thread (unbleached, white or in colors, as specified).

IVA - Soft finish.
IVB - Glazed finish.

1.2.2 Sizes. Sizes are designated by ticket numbers and approximate tex No. and shall be in accordance with tables I through V.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-50 - Packaging and Packing of Thread for Domestic and Overseas Shipment.

Federal Standards:

FED-STD-191 - Textile Test Methods.
FED-STD-751 - Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

The Color Association of the United States, Inc.

Department of Defense Standard Shades for Sewing Thread 1968

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, NY 10016.)

American Society for Testing and Materials (ASTM)

D 861 - Standard Recommended Practice for Use of the Tex System to Designate Linear Density of Fibers, Yarn Intermediates, Yarns, and Other Textile Materials.

E 380 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Government and supply purchases. The requirements specified in 3.8, 3.9, 3.10, section 5, and related provisions of section 4 apply only to thread purchased directly by the Government. All other requirements apply both to thread purchased by a supplier as a component for an end item and to thread purchased directly by the Government.

3.2 Standard sample. When a standard sample is established (see 6.2), the thread shall match the standard for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced.

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3.3 Material. The thread shall be made of combed cotton (see 4.2.4).

3.4 Color. The color (shade) shall be as specified in the applicable end item specification or in the contract or order (see 6.1). Shade designations by letter (e.g., Olive Drab S-1) and related cable numbers (e.g., CA 66022) refer to the DOD Standard Shades for Sewing Threads 1968 (see 2.2).

3.4.1 Matching. The shade of the end item thread, after removal of finish when applicable, shall match the specified applicable approved shade standard (see 6.2) when tested as specified in 4.2.4. Unbleached thread shall not be matched for shade.

3.4.2 Sulfur content. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished thread shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.4. When a standard sample is not available, the dyed and finished thread shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.4.

3.4.3 Colorfastness. The colorfastness of dyed finished thread shall be as specified in the applicable end-item specification or as set forth in the contract or order (see 6.1), except that when requirements are not stated or referenced to a standard sample for colorfastness, the thread shall show "good" colorfastness to laundering (after 3 cycles), light and perspiration. When a standard sample for colorfastness is referenced, the thread shall be equal to or better than the standard sample for the specified colorfastness properties or (when specific properties are not specified) for colorfastness to laundering (after 3 cycles), light and perspiration. Test for colorfastness for other characteristics such as colorfastness to wet drycleaning and weathering, when specified, shall be as specified in 4.2.4, as applicable.

3.5 Finish. The thread shall have a soft, glazed or mercerized soft finish as indicated for the applicable type and specified in the applicable end item specification or procurement document. The finish shall conform to the applicable requirements of 3.5.1 through 3.5.5 when tested in accordance with 4.2.4.

3.5.1 Soft finish. Soft finish thread shall have no consistency or luster other than that inherent to cotton material and imparted by the approved sewing finishes.

3.5.2 Glazed finish. Glazed finish thread shall be polished and lustrous, and shall have a smooth dressed surface.

3.5.3 Mercerized soft finish. Mercerized thread shall have been subjected to a mercerizing process, and shall have a soft finish. The 3-ply thread shall be singed (see 4.2.4).

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3.5.4 Finishing materials. No chemical finishes other than those specified shall be used without prior approval of the contracting officer. No finish or treatment shall be applied for the purpose of increasing breaking strength (see 4.2.4).

3.5.5 Sewing finishes. Unless prior approval is obtained from the contracting officer (see 4.2.4) only beeswax, paraffin, petroleum waxes, and dimethylpolysiloxanes or combinations thereof commonly used shall be permitted as sewing finishes for soft thread and mercerized thread. Any of the above in addition to starch shall be permitted for glazed finish thread.

3.6 Direction of twist. Unless otherwise specified (see 6.1), the direction of final twist shall be "Z" twist for all thread except the following which shall be "S" twist, determined as specified in 4.2.4.

Ticket No. 70 and 3-ply, of type IA, used for stitch type 606 of FED-STD-751.
Ticket No. 30, 2-ply of type IC when used as cover thread for stitch types 606 or 607 and as looper thread for stitch type 406.
Ticket No. 40, 4-ply of types IA and IB when used for buttonholes.
All sizes of types IVA and IVB when used for inseaming and as the bobbin of sole edge stitching in footwear.

3.7 Physical requirements. The finished thread shall conform to the applicable requirements specified in tables I through V for number of plies, length per pound, and breaking strength when tested as specified in 4.2.4.

3.7.1 Glazed thread. The minimum yards per pound shall be not less than 90 percent of the minimums specified.

TABLE I. Machine thread types IA1, IA2, and IA3 (soft) types IB1, IB2, IB3 (glazed)

Ticket No.	Appr. tex (g/km)	Ply	Length per pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
70	31	2	14,065 - 17,945	1.2	5.3
90	25	2	17,946 - 21,840	0.9	4.0
140 1/	19	2	23,000 - 28,750	1.1	4.9
20	86	3	5,040 - 6,440	3.8	16.9
30	67	3	6,441 - 8,400	3.0	13.3
40	54	3	8,401 - 10,080	2.1	9.3
50	42	3	10,081 - 13,440	1.9	8.5
70	33	3	13,441 - 16,800	1.5	6.7
70 2/	30	3	15,120 - 17,900	1.6	7.1

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TABLE I. Machine thread types IA1, IA2, and IA3 (soft) types IB1, IB2, IB3 (glazed) (cont'd)

Ticket No.	Appr. tex (g/km)	Ply	Length per pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
12	148	4	2,811 - 3,880	7.5	33.0
16	111	4	3,881 - 5,040	5.5	24.5
20	92	4	5,041 - 5,720	4.7	20.9
24	76	4	5,721 - 7,275	4.1	18.2
30	63	4	7,276 - 8,535	3.2	14.2
36	55	4	8,536 - 9,405	2.7	12.0
40	47	4	9,406 - 11,835	2.5	11.1
50	39	4	11,836 - 13,316	2.0	8.9
40	46	2/3	9,500 - 12,185	2.3	10.2

1/ Made from high grade cotton for hem stitching and handkerchiefs.

2/ Long staple cotton yarns "S" twist for 606 stitch only.

TABLE II. Machine thread types IC1 and IC2 (mercerized)

Ticket No.	Appr. tex (g/km)	Ply	Length per pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
30	39	2	11,001 - 14,400	1.6	7.1
36	32	2	14,401 - 16,800	1.3	5.8
C	72	3	6,010 - 7,760	4.0	17.8
B	58	3	7,761 - 9,215	3.2	14.1
A	51	3	9,216 - 10,280	2.8	12.4
0	44	3	10,281 - 12,220	2.6	11.6
00	36	3	12,221 - 15,520	2.1	9.3
000	27	3	15,521 - 21,340	1.7	7.6

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TABLE III. Basting thread types IIA1 and IIA2 (glazed)

Ticket No.	Appr. tex (g/km)	Ply	Length per pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
30	62	3	7,531 - 8,472	2.5	11.1

TABLE IV. Heavy thread type IIIA (soft) type IIIB (glazed)

Ticket No.	Appr. tex (g/km)	Ply	Length per pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
8	233	3	2,001 - 2,250	7.0	31.0
8	283	4	1,501 - 2,000	11.0	49.0
9	255	4	1,831 - 2,060	10.0	44,0
10	187	3	2,501 - 2,815	6.5	29.0
10	225	4	1,901 - 2,500	9.5	42.0
10	292	5	1,501 - 1,900	12.5	56.0
10	358	6	1,275 - 1,500	15.5	69.0

TABLE V. Shoe thread type IVA (soft) type IVB (glazed)

Ticket No.	Appr. tex (g/km)	Ply	Length stength pound, (yds.) min. - max.	Minimum breaking strength Pounds-Force (lbf)	Newton (N)
8	292	4	1,601 - 1,800	15	67
8	352	5	1,221 - 1,600	20	89
8	447	6	1,001 - 1,220	24	107
8	530	7	871 - 1,000	28	124
8	608	8	761 - 870	32	142
8	691	9	676 - 760	36	160
8	772	10	611 - 675	40	178
8	855	11	551 - 610	44	196
8	944	12	501 - 550	48	213

3.8 Knots and splices. Types I and II thread shall average not more than one full thread knot or one splice in the single yarn per 1,000 yards. Types III and IV thread shall contain no full thread knots. Type III, and sizes 8/4, 8/5 and 8/6 of type IV thread shall average no more than one splice in the single yarn per 500 yards. Sizes 8/7, 8/8, 8/9, 8/10, 8/11, 8/12 of type IV thread shall average no more than one splice in the single yarn per 250 yards.

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3.9 Put-up. Unless otherwise specified, the thread shall be put up on a nominal yardage or weight per holder basis on commercial spools, cones, or tubes (hereinafter referred to as holders) as specified in the contract or order in accordance with applicable schedules of tables VI through X (see 6.1). The average length or weight (as applicable per holder shall be not less than the specified nominal amount minus 3.0 percent tolerance when examined (on a 10-holder basis) as specified in 4.2.2. The thread shall be put up in one continuous length per holder and shall be so wound that each turn and layer is free from entanglement.

3.9.1 Ready made lockstitch bobbins. When put up on lockstitch bobbins, the style of bobbin and length of thread thereon shall be specified in the contract or order (see 6.1).

TABLE VI. Put-up for types IA, IB, IIA, IIIA, and IIIB on small double-end plastic spools

Ticket No.	Length, yards and meters
All	As specified

TABLE VII. Put-up for type IA on cones

Ticket No.	Approximate tex	Length					
		Yards			Meters		
20	92	6,000			5,500		
30	67	6,000	or	9,600	5,500	or	8,800
40	47	6,000	or	12,000	5,500	or	11,000
50	42	6,000	or	12,000	5,500	or	11,000
70,2	36	6,000	or	12,000	5,500	or	11,000
70,3	31	6,000	or	12,000	5,500	or	11,000
90	25	12,000	or	24,000	11,000	or	22,000
140	19	12,000	or	24,000	11,000	or	22,000

TABLE VIII. Put-up for types IB and IIA on single headed spools

Ticket No.	Approximate tex	Ply	Length	
			Yards	Meters
All	All	2	12,000	11,000
20	86	3	7,200	6,600
30	67	3	9,600	8,800
40	54	3	12,000	11,000
50	42	3	12,000	11,000
12	148	4	3,600	3,300
16	111	4	4,800	4,400
20 through 50	39 through 92	4	6,000	5,500

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TABLE IX. Put-up for types IC1 and IC2 on spools and cones

Ticket No.	Approximate tex	Cones length		Spools length		Cones net weight	
		Yards	Meters	Yards	Meters	Ounces	Grams
30	39	-	-	-	-	8	225
36	33	-	-	-	-	8	225
C	72	6000	5500	400	365	-	-
B	58	6000	5500	500	457	-	-
A	51	6000	5500	600	548	-	-
0	44	6000	5500	700	640	-	-
00	36	6000	5500	800	731	-	-
000	27	6000	5500	1000	941	-	-

TABLE X. Put-up for types IIIA, IIIB and IVA and IVB on straight tubes 1/

Ticket No.	Type	Nominal new weight	
		Ounces	Grams
All	IIIA	16	450
All	IIIB	16	450
All	IVA	16 or 32	450 or 900
All	IVB	16 or 32	450 or 900

1/ Unless otherwise specified, plastic tubes are prohibited.

3.10 Identification marking. Except when commercial identification markings are specified, each holder except the ready made lockstitch bobbins, shall have a label attached in such a manner as to remain in place and be clearly legible until all thread has been removed. In the case of ready made lockstitch bobbins, the label should be put on the outside of the unit pack (gross pack). The label shall be printed with the information as specified below:

Stock number
Length, yards and meters or nominal net weight ounces and grams
Direction of twist
Color
Ply
Ticket number
Approximate tex
Type
Nomenclature

3.10.1 Dating (type IV only). Marking for type IV shall also include date (month and year) of manufacture.

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3.11 Workmanship. The finished thread shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 End item examination.

4.2.1.1 Thread on the holder. The thread shall be examined on the holder and the visual defects listed in table XI shall be counted. The sample unit shall be one holder except for ready made bobbins shall be one gross. The lot size shall be expressed in units for one holder each except for ready made bobbins shall be expressed in units of one gross. The AQL shall be 4.0 defects per 100 units and the inspection level shall be S-3 of MIL-STD-105.

TABLE XI. Visual defects of thread on holders

Examine	Defects
Identification label	Missing, incorrect, incomplete, illegible or insecurely attached.
Type of holder	Other than specified.
Surface condition	Loose ply. Cut, tear, chafe, slip, affecting strength of thread or interfering with easy location of end and initial unwinding.

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TABLE XI. Visual defects of thread on holders (cont'd)

Examine	Defects
Color	Other than specified. Uneven, apparent on succesive layers or end to end.
Cleanness	Dirt, spot or stain clearly noticeable.
Finish	Other than specified. Not lustrous (glazed and mercerized finishes).

4.2.1.2 Thread as unwound. The thread shall be examined by unwinding and defects listed in table XII shall be counted regardless of their proximity to each other. The sample unit shall be one holder except for ready made bobbins shall be one gross. The lot size shall be expressed in units of one holder each, except for ready made bobbins shall be expressed in units of one gross. The AQL shall be 4.0 defects for 100 units and the inspection level shall be S-3 of MIL-STD-105.

TABLE XII. Defects of thread as unwound

Examine	Defects
Continuous length	Not in continuous lengths.
Knots and splices	Averaging more than the specified maximum of full thread knots or splices: types I and II thread, more than one full thread knot or one splice in the single yarn per 1,000 yards; type III, and sizes 8/4, 8/5 and 8/6 of type IV thread more than one splice in the single yarn per 500 yards, Sizes 8/7, 8/8, 8/9, 8/10, 8/11, 8/12 of type IV thread - no more than one splice in the single yarn per 250 yards.
Winding	Improperly or not firmly wound, resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of thread.
Tackiness or adhesion	Strands of thread adhere to each other or to holder affecting unwinding tension.
Thread uniformity	Uneven, lumpy or thin spots.

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4.2.2 Examination for length and net weight. The sample unit shall be 10 wound holders, taken at random from the lot under test.

4.2.2.1 Average length per holder. Ten holders of each lot shall be measured individually. Measurements shall be made over an accurate measuring clock under normal winding tension. The values obtained shall be averaged, and this average shall represent the length per holder in the lot. The lot shall be unacceptable if the length per holder is less than the amount specified minus the 3.0 percent tolerance (see 3.9).

4.2.2.2 Average net weight per holder. Ten holders of each lot shall be weighed individually. The weights of thread obtained shall be averaged and this average shall represent the net weight of thread per holder in the lot. The lot shall be unacceptable if the net weight per holder is less than the amount specified minus the 3.0 percent tolerance (see 3.9).

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-50 to determine whether packaging, packing and marking comply with the section 5 requirements.

4.2.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table XIII shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes. All test reports shall contain the individual values utilized in expressing the final result. The AQL for each characteristic shall be 6.5 test failures per 100 units and the inspection level shall be S-1. The sample unit shall be one wound holder. (For ready-wound bobbins, a sufficient number shall be utilized to total 350 yards.) The lot size shall be expressed in units of one holder, or one gross of ready-wound bobbins.

TABLE XIII. Test methods

Characteristic	Requirement paragraph	Test method
Combed cotton fiber identification	3.3	1200 1/
Shade match	3.4.1	4.2.4.1 and 9010
Presence of labile sulfur	3.4.2	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.4.3	5610 2/ 3/
Light	3.4.3	5660
Wet dry cleaning	3.4.3	5622
Perspiration	3.4.3	5680
Weathering	3.4.3	5671
Finish:		

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TABLE XIII. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Specified type	3.5	Visual
Mercerization (type IC)	3.5.3	Microsopic examination for swelling of fibers 1/
Singed (type IC - 3 ply)	3.5.3	Visual 1/
Materials limitations	3.5.4, 3.5.5	1/
Direction of final twist	3.6	4050
Number of plies	3.7	Visual 4/
Length per pound	3.7	4010 5/
Breaking strength	3.7	4100 6/

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens must be dried after each of the 3 laundering cycles.

3/ On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.

4/ One determination and the result reported as "pass" or "fail".

5/ In case of dispute, the length per pound of the thread shall be determined on the thread after removal of finish in accordance with procedure specified in 4.2.4.1.1.

6/ Except that five determinations shall be made on each sample unit.

4.2.4.1 Shade evaluation.

4.2.4.1.1 Removal of finish. Before evaluation for shade matching, the sample of thread shall be wet drycleaned in accordance with Method 5622 (disregarding references to the standard sample). Excess solvent shall be removed by centrifuging or wringing. The sample shall then be rinsed in distilled water at 120° to 160°F. and dried at a temperature not exceeding 180°F. The dried sample shall then be conditioned for a minimum of 4 hours prior to evaluation for shade match according to the procedure of Method 9010.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

5.1.1 Levels A, B, and C. The thread put-up as specified shall be packaged in accordance with the applicable requirements of PPP-P-50.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Levels A, B, and C. The thread shall be packed in accordance with the applicable requirements of PPP-P-50.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-50.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, finish, ticket number and ply, and nomenclature of thread required (see 1.2).
(c) Color (shade) required (see 3.4).
(d) Colorfastness properties required (see 3.4.3).
(e) Direction of twist (see 3.6).
(f) Put-up required (see 3.9).
(g) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).
(h) Style of bobbin and length of thread for ready made lockstitch bobbins (see 3.9.1).

6.2 Standard sample. For Military shades, the standards for shade reference are those contained in the Standard Color Card for the Official Standardizied Shades of Sewing Threads, 1968 (see 2.2 and 3.4). For other shades, address the procuring office issuing the invitation for bids or request for proposal. Standard samples frequently used by the military (directly or for components of end items) have been made available to normal suppliers as basic reference for the establishment of the supplier's secondary standards conforming to the characteristics of the official standards. The basic shade standards are those appearing in the DOD Color Card. Prime thread manufacturers who inadvertently have not been furnished standards for establishment of their colorfastness standards in accordance with this specification may obtain samples for this purpose from the U. S. Army Natick Research and Development Command, Natick, MA 01760, ATTN: DRXNM-VT.

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6.3 Colorfastness. Normally, colorfastness requirements for thread are to be specified in keeping with those specified for the basic material into which the thread will be sewed (see 3.4.3).

Typical examples are:

Use	Colorfastness
Thread for sewing wool clothing which is drycleaned.	Good fastness to wet drycleaning, perspiration, and light.
Thread for sewing wool clothing which is laundered.	Good fastness to laundering, perspiration, and light.
Thread for sewing cotton clothing which is laundered.	Good fastness to laundering (after 3 cycles), perspiration and light.
Thread for sewing footwear.	No colorfastness test apply.
Thread for sewing tentage.	Good fastness to weathering.
Thread for sewing equipage.	Good fastness to weathering and in special cases to laundering (after 3 cycles).

6.4 Dye formulations . Suggested but not mandatory formulations for the shades indicated are as follows:

Khaki Shade P-1-CA 66019
Vat Brown G. PR 380
Vat Black 27, C.I. 69005
Vat Yellow Brown 3G, PR 125

Olive Drab Shade S-1-CA 66022
Vat Black 27, C.I. 69005
Vat Green 3, C.I. 69500/1
Vat Yellow Brown 3G, PR 125
Vat Brown G, (as required) PR 380

Green Shade E-CA 66034
Vat Green 3, C.I. 69500/1
Vat Black 27, C.I. 69005
Vat Blue 18, C.I. 59815

6.5 Size. In order to eventially adopt one uniform sizing method for threads of all types, the tex system has been selected. The tex number is defined as the weight in grams of 1000 meters (1 km) of yarn or other textile strand in accordance with ASTM: D 861. The approximate tex number is shown alongside of the ticket number.

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6.6 Physical requirements. In view of the proposal to metricate all specifications and establish the use of SI units (ASTM: E 380) the breaking strength requirements are shown in Newtons (force unit SI system) as well as the standard pounds force.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD, EA
Navy - OS
Air Force - 99
DSA - CT

User activities:

Navy - SH, MC

Preparing activity:

Army - GL

Civil Agency Coordinating Activity:

GSA - FSS
HEW - HSM
VA - DMS

Project No. 8310-0098

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